

Correcting the 2nd and 3rd 90s through Laboratory Surveillance

Amoy Douse-Mullings, Selena Lewis, Akeem Whitehead, Omar Elliott, Christine Ehikhamenor,
Kereann Nelson, Nicola Skyers, Rahanah Khan-Francis
HIV/STI/TB Unit, Health Promotion and Protection Branch, Ministry of Health & Wellness,
Jamaica

Abstract

Background

Globally healthcare systems continue to experience significant impact by the Human Immunodeficiency Virus (HIV) epidemic. While Jamaica has made notable progress in achieving the first 90 of the UNAIDS 90-90-90 targets, challenges persist in achieving the second and third 90s, specifically in retaining persons living with HIV (PLHIV) in care and achieving viral suppression. Loss to follow-up (LTFU) remains a critical barrier, with significant gaps in national treatment cascades. Laboratory surveillance offers a potential solution to identify and track PLHIV categorized as LTFU by definition, but are accessing care at private and civil society organizations and not being documented as such in the Treatment Services Information System (TSIS). This allows for more accurate estimations of care engagement and viral suppression.

Method

A retrospective cohort study was conducted in March 2024 using data from the TSIS and District DISA Lab, an information management system utilized by the National Public Health Laboratory (NPHL). A total of 19,391 PLHIV linked to care in TSIS2 were analyzed, of which 3,315 were classified as LTFU. Data on viral load testing and suppression were manually compared across various healthcare providers, and statistical analyses were performed to assess trends in care engagement and viral suppression.

Results

The study revealed that 54% of the LTFU cohort were female, and the age group 30-39 years accounted for the highest proportion (29%). Approximately 4% of the LTFU cohort had valid viral load results, with 70% achieving viral suppression. Public healthcare providers accounted for 92% of LTFU cases, while NGOs and private providers contributed 6% and 2%, respectively. Viral

suppression rates among those with valid viral loads were highest in older age groups, with an overall suppression rate of 73%.

Conclusion

Significant gaps in the care of PLHIV categorized as LTFU was identified through laboratory surveillance. Despite being classified as out of care, many PLHIV had valid viral load results and achieved viral suppression. Addressing LTFU through enhanced surveillance could lead to improved estimates of viral suppression and targeted interventions, supporting Jamaica's progress toward the UNAIDS 95-95-95 targets.

Keywords: HIV, loss to follow-up, viral load, viral suppression.

Introduction

The epidemic of the Human Immunodeficiency Virus (HIV) continues to wreak substantial impact on public health systems globally (1). Since the first recorded case of HIV in 1959 (2) there has been significant progress in the fight against the disease. Despite the advancements in medicine and public health research, millions of people living with HIV (PLHIV) continue to face barriers relating to the accessibility of effective and consistent HIV related care. According to the Joint United Nations Programme on HIV/AIDS (UNAIDS), in 2023 alone 39.9 million people were estimated to be living with the disease globally, 1.3 million new cases were recorded, and 630,000 lives were lost due to AIDS related illnesses (3). Since 2021, UNAIDS has set new targets (95-95-95) to eliminate the AIDS epidemic by 2030 (4). However, the 2014, 90-90-90 targets aim to achieve the following goals: 90 % of PLHIV know their status, 90% of diagnosed PLHIV are retained on antiretroviral therapy (ART) and 90% of those on ART are achieving a status of being virally suppressed by 2025.

In the case of Jamaica, the progress towards achieving the first 90 has been promising, the challenge lies with the 2nd and 3rd 90s (5) . A common challenge with hitting these targets is the issue of loss to follow up (LTFU), where PLHIV who have been diagnosed withdraw from regular treatment and monitoring for greater than 90 days (6). Universally, there are several factors such as inadequate healthcare systems, stigma, treatment fatigue and access to care that contributes to LTFU (7). This issue poses a significant obstacle in Jamaica as identified through the Treatment Services Information System 2.0 (TSIS). TSIS is utilized by the Ministry of Health and Wellness (MOHW) Jamaica to capture and track PLHIV data. The barrier of LTFU in understanding and addressing HIV treatment is evident when ascertaining whether these PLHIV are accessing care through other public healthcare systems, private or other informal entities.

Critical to achieving the global target of ending the HIV epidemic are the 2nd and 3rd 90s. The impact of PLHIV that are LTFU presents gaps in national treatment cascades, therefore undermining treatment engagement and viral suppression as well as impeding the effects of targeted interventions in PLHIV care (8). Laboratory surveillance is a method that can be used to rectify this problem, it serves as a tool for tracking the viral loads of PLHIV identified as being LTFU, with the potential to pinpoint PLHIV who are currently receiving treatment outside of

formal care (9). This approach might possess the potential to enable more accurate estimates of national treatment cascades leading to more informed interventions to combat LTFU and address rates of viral suppression. This study aims to understand the role of laboratory surveillance in tracking HIV patients who are LTFU and ascertaining whether these PLHIV are still receiving treatment. We focused on whether PLHIV who are loss to follow-up (LTFU) in TSIS are truly out of care and the potential gains for the national treatment cascade should the PLHIV identified as being LTFU are accessing care elsewhere.

Method and Materials

Study Design

This study utilized the retrospective cohort design, conducted in March 2024. For this research quantitative data was extracted from TSIS, which hosts information for PLHIV accessing care public and selected private facilities under the public/private partnership in Jamaica. At the time, there were 19,391 PLHIV linked to care in TSIS2 of which 3,315 were classified as LTFU based on the predefined inclusion criteria.

Definition of key terms

Lost to follow up

PLHIV lost to follow up are defined as individuals who have defaulted from care for ≥ 91 days from their last scheduled clinic appointment. This was the primary criterion for the inclusion of the LTFU cohort. The aims of the research necessitated the extraction of the LTFU cohort from TSIS which served as the cohort for statistical analysis.

Valid Suppression

Viral load and is less than 1000 copies per ml.

Valid Viral Load

Viral load result dated within the last 12months.

Invalid Viral Load

Viral load result dated greater than 12 months.

Source and Collection of Data

TSIS2 and DISA Lab which houses all viral load samples processed nationally served as the main sources of data for this study. The dataset containing the LTFU population was pulled from the TSIS2 database, after which DISA Lab was engaged to ascertain if the PLHIV LTFU cohort had a recent viral load. The variables for the comparison between both sources were first name, last name, date of birth, medical record number, last viral load test date, last viral load test result, and facility that requested the viral load test.

Statistical Analysis

We used Excel for data analysis and presentation. First, we compared the PLHIV that were successfully linked to care versus those from that population who became LTFU. This comparison was disaggregated by sex and involved an inspection of the distribution of this population across service providers. The LTFU population was further analyzed, disaggregated by sex and gender. We examined the viral load results and suppression status for the LTFU cohort. This included a detailed analysis of valid viral load results. The valid viral load results were then further disaggregated by sex and age. A focused analysis was performed on the LTFU individuals with valid viral load results categorized by service provider. The suppression status and age group of these PLHIV were also analyzed followed by an assessment of the impact on the National Treatment Cascade for the month of September. Finally, an analysis was done of the LTFU cohort with invalid valid load results, the distribution of these individuals by service provider and a summary of our key findings compiled.

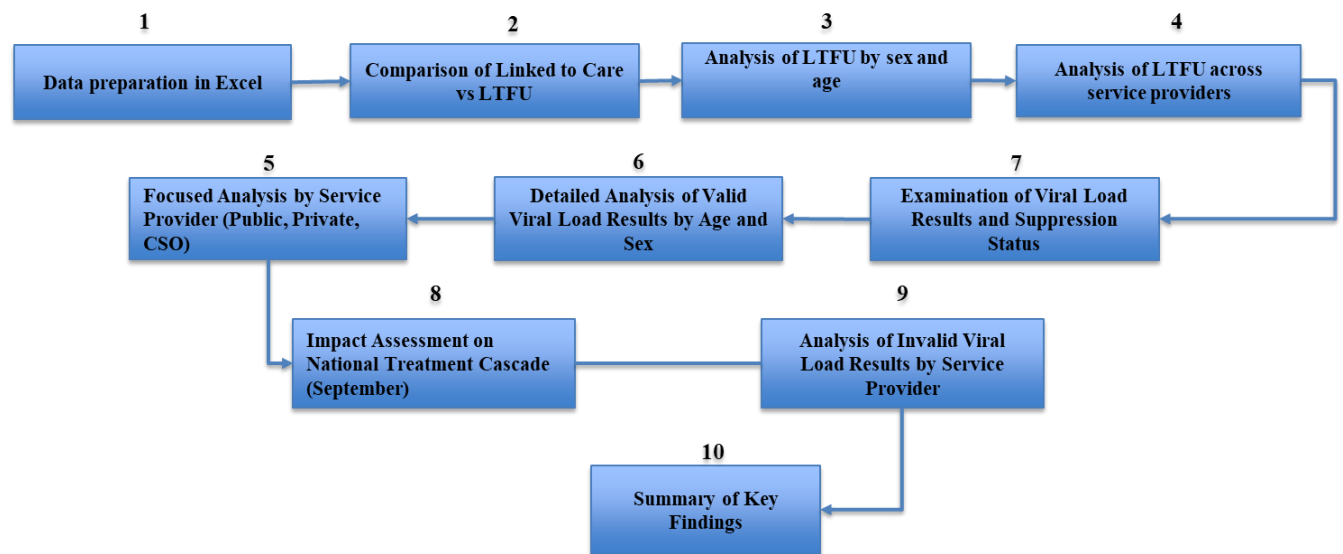


Figure 1. Steps in Statistical analysis.

Results

Sex and Age Disaggregation of LTFU Cohort

The analysis between linked to care and LTFU revealed trends across age groups (Table 1). For males, the proportion LTFU relative to the total population showed significant variations; the age group 30-39 years reflected the highest proportion of 23%, followed by age group 20-29 with 22%. For females, the proportion of individuals LTFU relative to the total population of females climaxes at age group 20-29 with 28% of females LTFU, followed by the age group 30-39 with 23%.

In comparing both sexes LTFU versus linked, males showed a slightly higher proportion of LTFU in the youngest age groups when compared to their female counterparts; males in the age groups 0-9 and 10-19 years recorded 17% and 14% respectively, while females in the same age groups showed 6% and 12% respectively. Conversely, in age groups >50 the LTFU was observed to be lower in females in comparison to males. Only 10% of females ages 50-59 were LTFU compared to 14% for males, while among those aged 60+, 9% of females were LTFU compared to 12% for males.

Table 1. Linked to Care Population Compared to LTFU Population Disaggregated by Sex

Age Group	Males Linked (n)	Males LTFU (n)	% Males LTFU / Male Population	Females Linked (n)	Females LTFU (n)	% Females LTFU / Female Population
0-9	35	6	17%	50	3	6%
10-19	125	17	14%	182	22	12%
20-29	1,049	228	22%	1,482	409	28%
30-39	1,704	392	23%	2,513	573	23%
40-49	2,084	382	18%	2,718	412	15%
50-59	2,237	314	14%	2,168	227	10%
60+	1,556	183	12%	1,452	135	9%
Missing	23	-	-	12	-	-
Total	8,813	1,522	23%	10,577	1,781	17%

A total of 3,315 persons were identified as LTFU (Table 2) within the linked to care cohort (19,391) retrieved from TSIS2. Of the LTFU population, 54% were females while the remaining 46% were males. The highest percentage of individuals LTFU was observed in age group 30-39 (965, 29.1%); females represented the greatest proportion of this age group 59%, and males represented 41% respectively. The 40-49 age group showed a slight decrease in the number of individuals LTFU (794, 24%). A total of 637 individuals were LTFU in the 20-29 age group. Females notably had more cases of LTFU (409), while males had 228 cases. A marked decrease in the number of LTFU individuals was observed in the 40-49 age group (794, 24%) and 50-59 age group (541, 16.3%), with the latter group showing a notable decline in females (227, 42%) compared to males (314, 58%). The elderly age group 60+ showed a lower total number of cases of LTFU (318, 96%), with more individuals being from the male gender (183, 58%) than the female gender (135, 42%). Only a total of 9 (0.2%) individuals made up the 0-9 age group.

Table 2. Lost to follow-up cohort disaggregated by age and sex.

Characteristics	Age Range	Male n (%)	Female n (%)	Total n (%)
Age Group	0-9	6 (0.2)	3 (0.1)	9 (0.2)
	10-19	17 (0.5)	22 (0.7)	39 (1.2)
	20-29	228 (6.9)	409 (12.3)	637 (19.2)
	30-39	392 (11.8)	573 (17.3)	965 (29.1)
	40-49	382 (11.5)	412 (12.4)	794 (24.0)
	50-59	314 (9.5)	227 (6.8)	541 (16.3)
	60+	183 (5.5)	135 (4.1)	318 (9.6)
Missing	-	-	-	12 (0.4)
Total		1,522 (46.0)	1,781 (54.0)	3,315 (100.0)

Retention and Viral load test status among LTFU Cohort

The majority of the LTFU cohort was receiving care in public healthcare facilities (3,045, 92%). In contrast, non-governmental organizations (NGOs) and private care accounted for 6% (193 individuals) and 2 % (77 individuals) respectively (Figure 2). Of the 3,315 the vast majority of 3,096 (94%) did not have a valid viral load test ,77 (2%) were never tested for viral load and only 142 (4%) had a valid viral load test result (Figure 3).

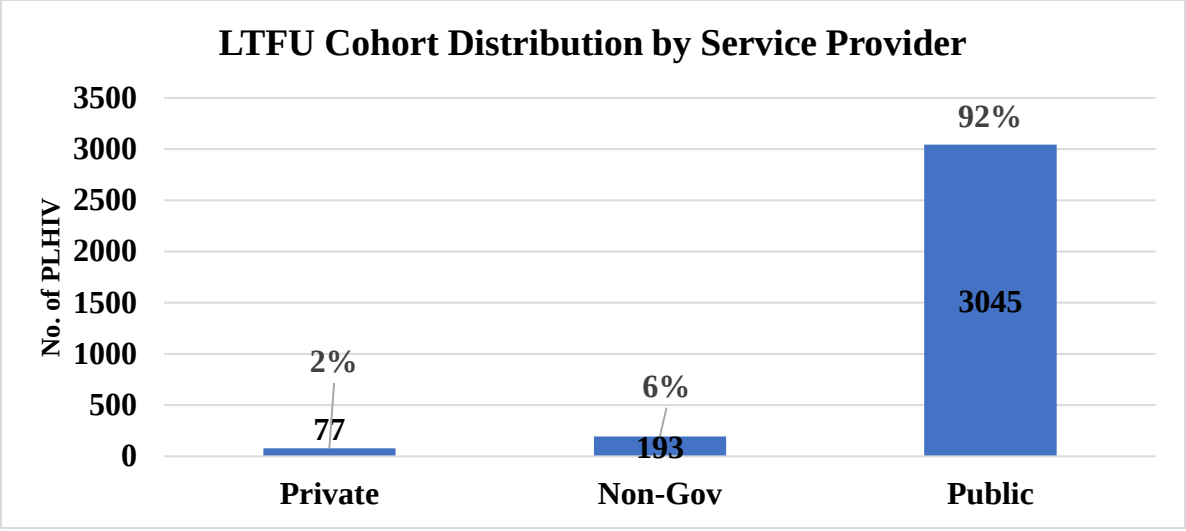


Figure 2. LTFU cohort distributed by service provider.

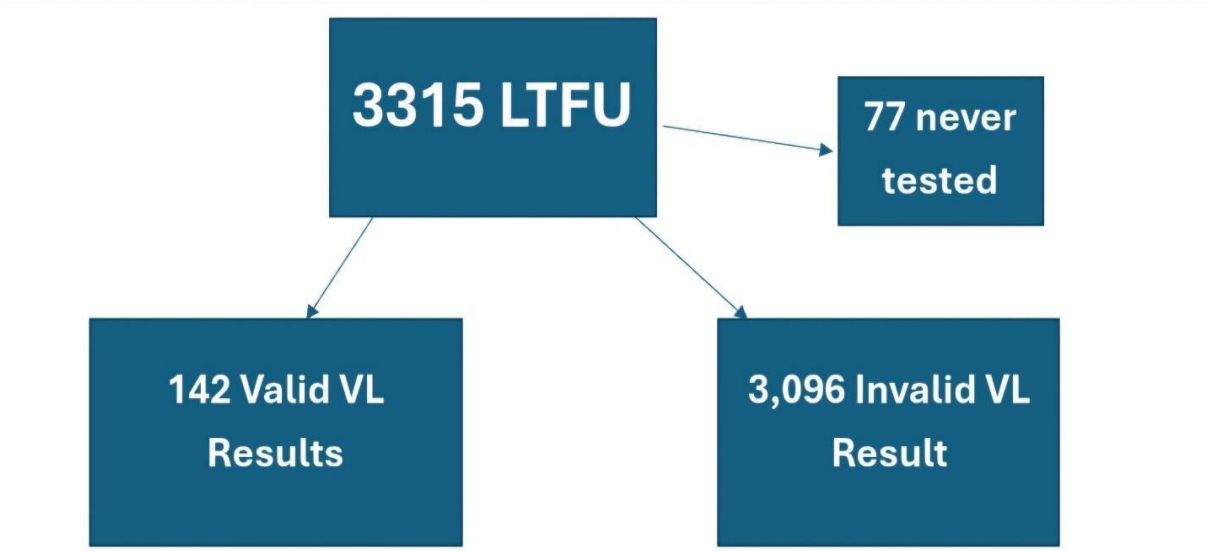


Figure 3. Breakdown of LTFU Cohort by Viral Load Testing Status.

Valid Viral Load Results among LTFU cohort

At the last recorded viral load test, 73% (2,423 individuals) were recorded as being virally suppressed, while 25% (815 individuals) were unsuppressed. The remaining 2% (77 individuals) were never tested (Figure 4). Of those with a valid viral load result, females constituted a significant proportion 85%, while males accounted for 15% (Figure 5). A closer look at the those with a valid viral load disaggregated by age and gender (Figure 6) reveals that among both sexes, PLHIV in the combined age group 30-49 years accounted for the majority (61% in males and 60% in females).

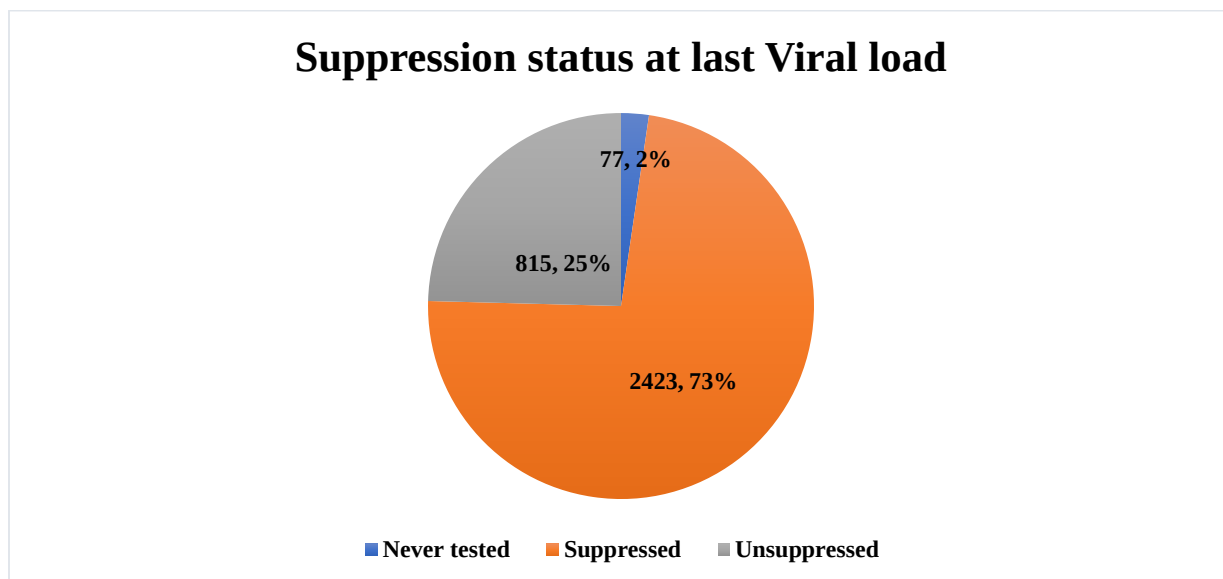


Figure 4. Suppression Status at Last Viral Load Test.

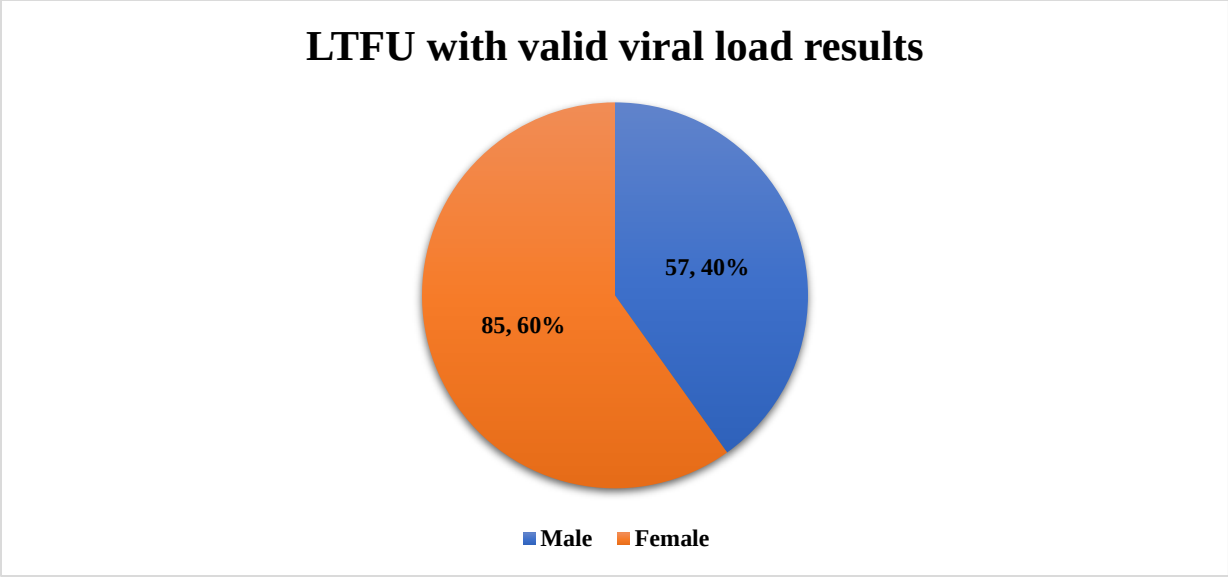


Figure 5. LTFU with Valid viral load results.

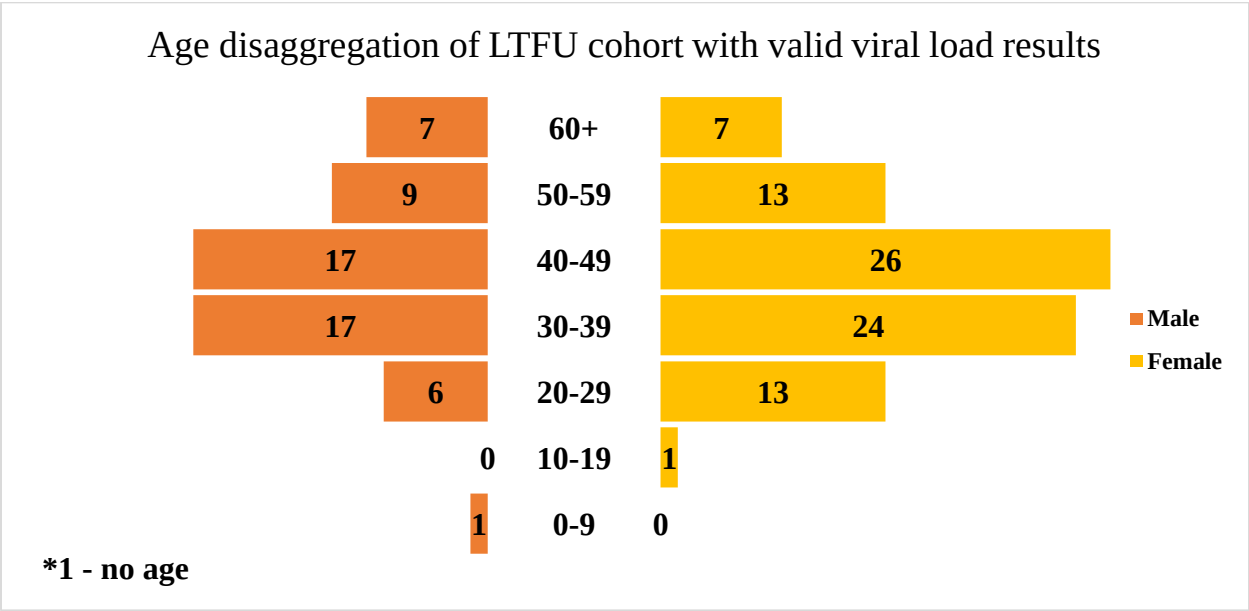


Figure 6. Age disaggregation of LTFU with valid viral load results.

Valid viral load results, viral suppression and national treatment cascade

As indicated by the data (Figure 7), the majority (67%) of valid viral loads results were accounted for by the public providers. The private sector contributes to 25%, while NGOs accounted for approximately 8% . Overall, a total of 100 (70%) individuals were virally suppressed whereas 42 were unsuppressed (30%) (Figure 8). A detailed look at the data specifically by age group showed that age groups 0-9 and 10-19 achieved a viral suppression rate of 100%. A slight decline in suppression rate was observed as age increased (20-39). A suppression rate of 68% (19 cases) was recorded for age 20-29, while a rate of 66% (41 cases) was recorded for individuals 30-39 years. The 40-49 age group had the highest suppression rate 72% (43 cases). Age groups 50-59 and 60+ years had suppression rates of 77% (22 cases) and 71 (14 cases) respectively (Figure 9). Our assessment of the impact of these individuals on the national treatment cascade revealed a 1% increase in viral load uptake and suppression (Figure 10).

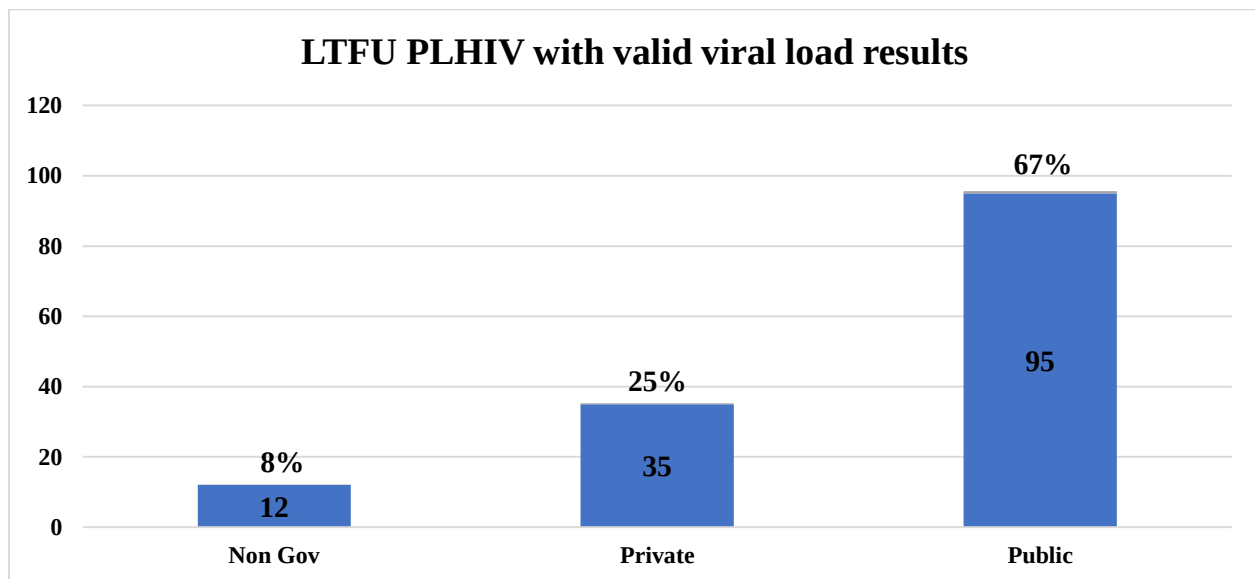


Figure 7. LTFU PLHIV with valid viral load by service provider.

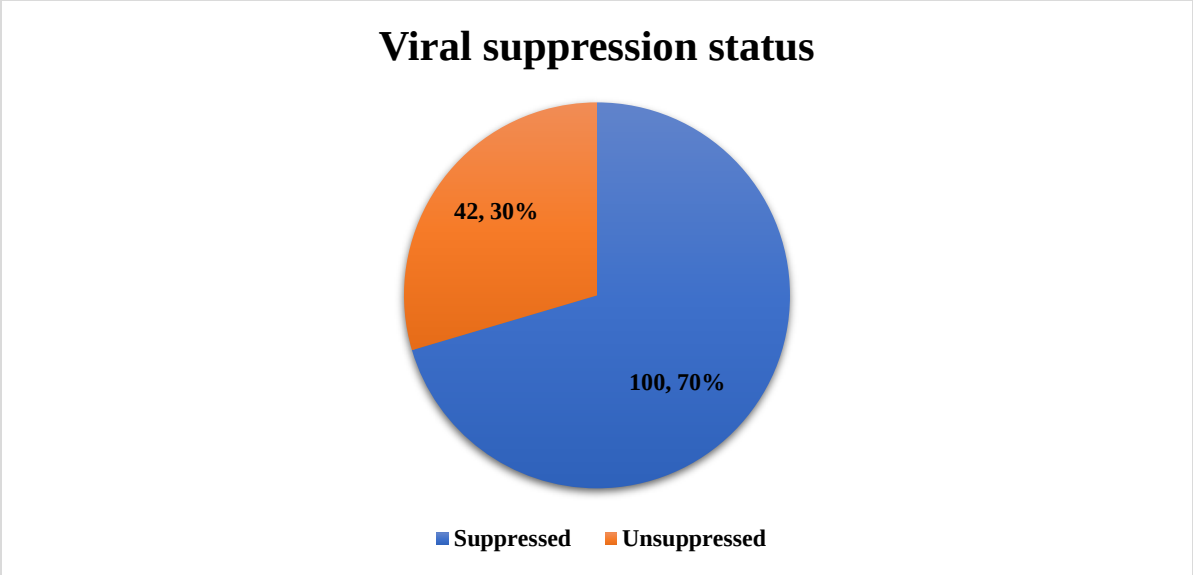


Figure 8. Viral suppression status

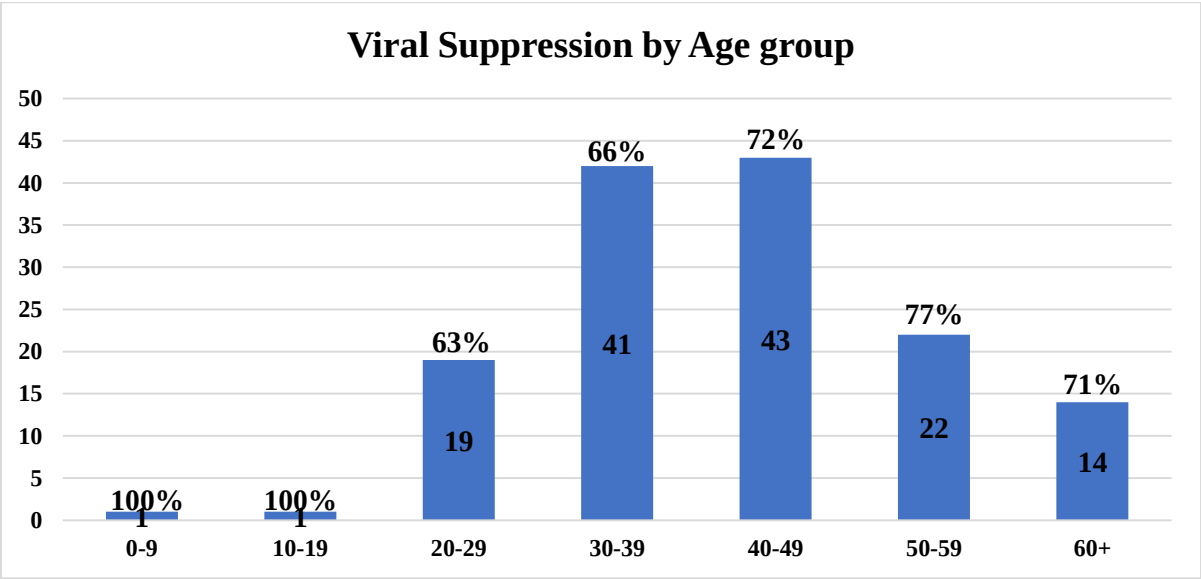


Figure 9. Viral Suppression by Age Group.

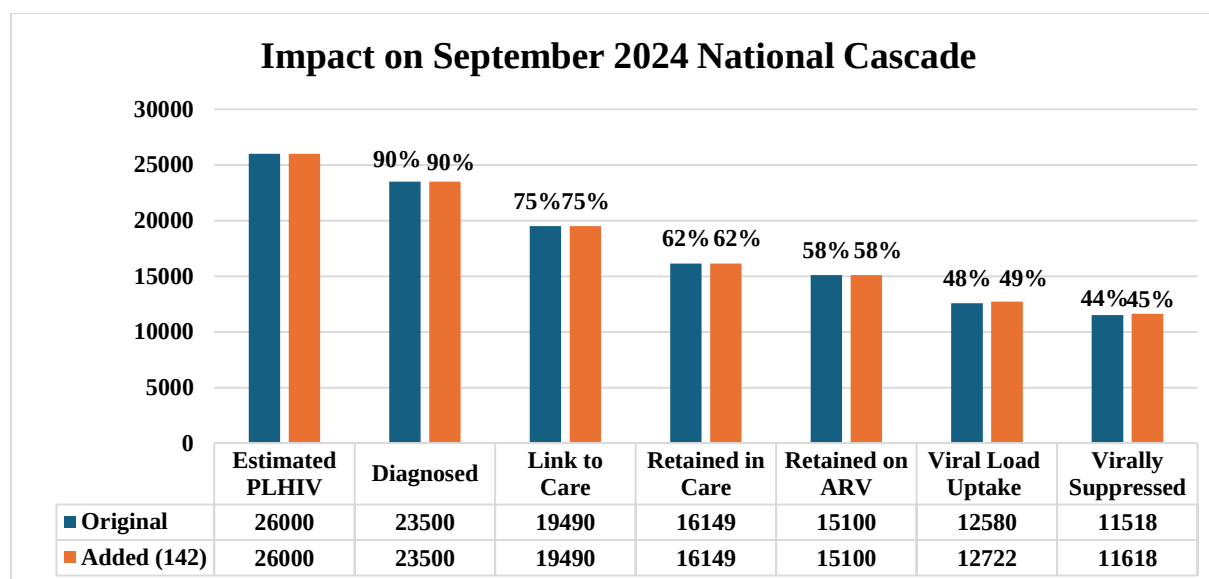


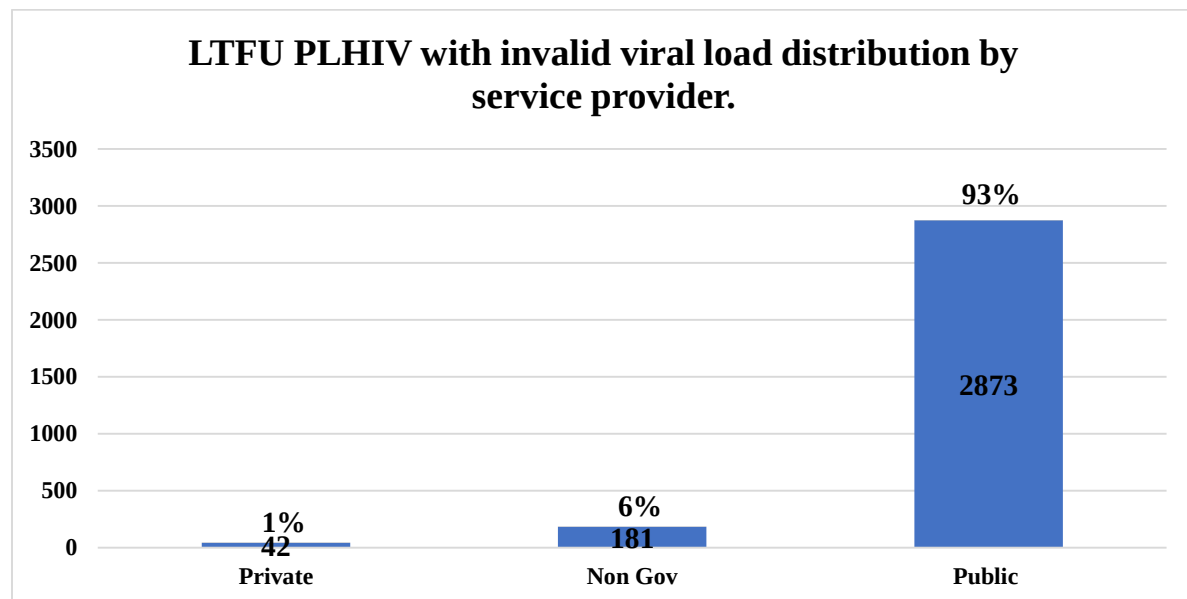
Figure 10. Impact of Additional Virally Suppressed Cases on September 2024 National Cascade

LTFU PLHIV with Invalid Viral Load Results

A total of 3,096 individuals made up the LTFU cohort with invalid viral load results; females accounted for 54% of the cohort in comparison to males 46%. Individuals in the age group 30-39 years constituted the largest group and accounted for 29.5%, followed but age groups 40-49 and 20-29 years with 24.4% and 19.3% respectively. Age groups 50 and above accounted for 25.2%, while those 0-19 years accounted for 1.3% of the cohort. There was a small proportion of the data that lacked age specific details (Table 3). Of the 3,096 LTFU individuals with invalid VL results, 2,873 (93%) were linked to public service providers. A small proportion of 181 (6%) individuals were linked to NGOs and the remaining 42 (1%) individuals were associated with private care providers (Figure 11).

Table 3. Age and Sex distribution of invalid result

Age Range	Male	Female	Number (%)
0-9	4	3	7 (0.2)
10-19	14	21	35 (1.1)
20-29	213	385	596 (19.3)
30-39	373	539	912 (29.5)
40-49	371	383	754 (24.4)
50-59	281	205	486 (15.7)
60+	172	121	293 (9.5)
Missing	-	-	11 (0.4)
Total	1,428	1,657	3,096

**Figure 11.** LTFU PLHIV with invalid viral load Distribution by Service Provider

Demographics and Clinical Characteristics of LTFU cohort

The analysis revealed key insights into the viral load testing and suppression rates of the LTFU cohort. A large proportion of cohort fell within the age range of 30-49 years, accounting for a total

of 53%. Additionally, this age group was predominant among valid viral load results and invalid viral load results 59% and 54% respectively. The disaggregation of sex revealed that more females (54%) were LTFU when compared to their male counterparts 46%. Females also represented the majority of the valid (60%) and invalid (54%) viral load test results. Overall, the cohort had a high viral suppression rate of 73%; those with a valid viral load was 70% and 78% at the last documented test for those who had an invalid viral load test result. There was a high coverage (98%) as it relates to viral load testing among those LTFU. Of these, 75 % of the valid results were reported from facilities in TSIS2 (Figure 12).

	LTFU PLHIV	Valid VL Results	Invalid VL results
Age distribution	30 – 49 years (53%)	30 – 49 years (59%)	30 – 49 years (54%)
Sex	More females were LTFU (54%)	More females had a valid VL result (60%)	More females had an invalid VL result (54%)
Viral Suppression Rates	73%	70% (higher in 40+ age group)	78% at last test
• High VL testing coverage among LTFU PLHIV- 98% • 75% of valid VL tests were from sites using TSIS			

Figure 12. Key Insights of Demographics and Clinical Characteristics of LTFU Cohort.

Discussion

This study evaluated the effect persons living with HIV (PLHIV) who are lost to follow-up with a valid viral load have on the viral suppression indicator on the 90-90-90 Jamaica national cascade. This led to a 1% increase in both the viral load uptake and viral suppression indicators.

Retention in care is directly associated with viral suppression (10,11,12). However, it was shown that a high proportion of PLHIV lost to follow with a valid viral load uptake up are virally suppressed. These findings indicate that though these patients may not be retained in care at their assigned treatment sites, access to treatment care and support is implied. Greater efforts are required to monitor and track patients across all treatment sites to more accurately estimate the retention and suppression pillars of Jamaica's national cascade.

The majority of PLHIV lost to follow-up (54%) were females and within the age group 30 -39 years (29%), attending public sites (92%). The age group remained consistent with previous studies. However, there was a higher proportion of females compared to males lost to follow-up compared to previous studies (13, 14). Gender based violence, intimate partner violence, stigma and discrimination, along with suboptimal patient provider communication (3) have been cited as barriers to access HIV treatment and care. Coordinated and integrated interventions are required to address these social and structural barriers in order to improve access and retention in care for women living with HIV.

Approximately 4% of the PLHIV lost to follow-up had valid viral load results during the period of March 2024, with the majority being females (60%) and within the age group 40 – 49 years (28%). While the age group corresponded with previous studies (14), the proportion of females compared to males was higher, as previous studies found a higher proportion of males to be adherent to the VL test/uptake (14).

Strengths

The study addresses critical gaps in achieving the 2nd and 3rd 90s of the UNAIDS targets, with a specific focus on retention and viral suppression in Jamaica, providing actionable insights for public health interventions. Secondly, the study disaggregates data by sex, age, and service

provider, allowing for a nuanced understanding of disparities and informing targeted interventions for different demographic groups. The study also emphasized the critical need for uninterrupted interoperability of TSIS and DISA Lab to closely monitor viral load uptake and suppression.

Limitations

This study is not without limitations. Data collection and comparison were done manually which can introduce human error, affecting data quality. The absence of a unique identifier for both TSIS and DISA made data comparison arduous and labour intensive, possibly resulting in cases that might have been missed. Additionally, fatalities as a treatment outcome depends on deaths being reported to the medical facility and recorded, as such some of the lost to follow-up could have died.

Recommendation

A more in depth investigation is required to understand the reasons for not updating TSIS at both public and private facilities, which was evident with the valid viral load tests results seen.

Conclusion

This study highlights the critical role of interoperability of TSIS and DISA Lab in addressing gaps in the care continuum for persons living with HIV (PLHIV) in Jamaica, particularly those categorized as lost to follow-up (LTFU). Despite being classified as out of care, a significant proportion of LTFU individuals had valid viral load results, with the majority achieving viral suppression. These findings suggest that some PLHIV may be accessing care outside formal public health systems, underscoring the importance of uninterrupted connectivity of TSIS and DISA Lab.

By identifying individuals who are virally suppressed despite being LTFU, targeted interventions can be developed to re-engage them in formal care and address data quality gaps in treatment systems. Addressing LTFU through such measures is essential for Jamaica to advance towards the UNAIDS 95-95-95 targets and ultimately eliminate the HIV epidemic.

This study underscores the need for improved data integration, streamlined processes, and expanded partnerships with private and non-governmental health providers to optimize the national response to HIV care and treatment.

Acknowledgement

We thank Sheldon Whorms and Andre Williams HIV/STI/TB unit MOHW along with Dr. Marlene Tapper and Dr. Kereann Nelson at the National Public Health laboratory.

Funding

This study received no specific grant from any funding agency.

References

1. Gayle HD, Hill GL. Global Impact of Human Immunodeficiency Virus and AIDS. *Clinical Microbiology Reviews* [Internet]. 2001 Apr 1;14(2):327–35. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC88977/>
2. Zhu T, Korber BT, Nahmias AJ, Hooper E, Sharp PM, Ho DD. An African HIV-1 sequence from 1959 and implications for the origin of the epidemic. *Nature* [Internet]. 1998;391(6667):594–7. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/9468138>
3. UNAIDS. Global HIV & AIDS Statistics — 2024 Fact Sheet [Internet]. UNAIDS. 2024. Available from: <https://www.unaids.org/en/resources/fact-sheet>
4. Frescura L, Godfrey-Faussett P, Feizzadeh A. A, El-Sadr W, Syarif O, Ghys PD. Achieving the 95 95 95 targets for all: A pathway to ending AIDS. Ambrose Z, editor. *PLOS ONE* [Internet]. 2022 Aug 4;17(8):e0272405. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9352102/>
5. Jamaica [Internet]. [Unaids.org](https://www.unaids.org/en/regionscountries/countries/jamaica). [cited 2024 Dec 31]. Available from: <https://www.unaids.org/en/regionscountries/countries/jamaica>
6. Endebu T, Taye G, Deressa W. Rate and predictors of loss to follow-up in HIV care in a low-resource setting: analyzing critical risk periods. *BMC Infectious Diseases*. 2024 Oct 18;24(1).
7. Mandawa MB, Mahiti GR. Factors Contributing to Loss to Follow-Up from HIV Care Among Men Living with HIV/AIDS in Kibaha District, Tanzania. *HIV/AIDS - Research and Palliative Care*. 2022 Nov;Volume 14:503–16.
8. Abebe Feyissa Amhare, Girma Garedew Goyomsa, Issa A. Investigating the follow-up discontinuation among people living with HIV in North Shoa Zone, Oromia, Ethiopia. *Frontiers in Public Health*. 2024 Oct 24;12.
9. Nsubuga P, Simbarashe Mabaya, Apollo T, Ngwarai Sithole, Komtenza B, Takura Matare, et al. Evaluation of the Zimbabwe HIV case surveillance pilot project, 2019. *Pan African Medical Journal* [Internet]. 2019 [cited 2024 Dec 30];37(1). Available from: <https://www.ajol.info/index.php/pamj/article/view/232284>
10. Yehia BR, Rebeiro PF, Althoff KN, Agwu AL, Horberg MA, Hasina Samji, et al. Impact of Age on Retention in Care and Viral Suppression. *Journal of Acquired Immune Deficiency Syndromes*. 2015 Apr 1;68(4):413–9.

11. Elashi BAY, Van Wyk BE. Factors associated with viral suppression among adolescents on antiretroviral therapy in Free State province, South Africa. *Southern African Journal of HIV Medicine*. 2022 Jun 13;23(1).
12. Crawford TN, Thornton A. Retention in Continuous Care and Sustained Viral Suppression. *Journal of the International Association of Providers of AIDS Care (JIAPAC)*. 2016 Nov 16;16(1):42–7.
13. Elgalib A, Shah S, Adil Al-Wahaibi, Zeyana Al-Habsi, Maha Al-Fouri, Lau R, et al. Retention in HIV care and factors associated with loss to follow-up in Oman: a countrywide study from the Middle East. *AIDS care*. 2021 Apr 29;34(5):568–74.
14. Hong SY, Winston A, Nicholus Mutenda, Ndapewa Hamunime, Roy T, Wanke C, et al. Predictors of loss to follow-up from HIV antiretroviral therapy in Namibia. *PloS one* [Internet]. 2022 Apr 14 [cited 2024 Apr 29];17(4):e0266438–8. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9009635/#:~:text=In%20Namibia%2C%20ART%20use%20by>
15. Lubega P, Nalugya SJ, Kimuli AN, Twinokusiima M, Khasalamwa M, Kyomugisa R, et al. Adherence to viral load testing guidelines, barriers, and associated factors among persons living with HIV on ART in Southwestern Uganda: a mixed-methods study. *BMC Public Health*. 2022 Jun 29;22(1).